

Immersive Motorcycle Visualization Using Motion Capture: A Proof of Concept for an Online Motorcycle Configurator

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Figure 1: Immersive Online Motorcycle Configurator: After the motorcycle has been configured, the vehicle can be experienced in a video game-like simulation.

ABSTRACT

The increasing adoption of online configurators in the automotive industry has transformed how customers personalize and visualize vehicles. In the motorcycle sector, however, existing configurators often lack dynamic and immersive user experiences, particularly when it comes to showcasing the interaction between the rider and the motorcycle. This paper presents a proof-of-concept prototype leveraging motion capture technology to create realistic interactions between a motorcycle and its rider within a virtual environment. Inspired by immersive features in racing games, we explore an optimized workflow for integrating motion capture data into a VR-based online configurator and examine challenges such as adapting motion animations to various motorcycle models. By incorporating a virtual avatar to represent the user, the prototype enhances the sense of presence and identification with the product, addressing a critical limitation of traditional configurators. We hypothesize that immersive, animated interactions between a motorcycle and its rider can enhance the user experience, fostering greater engagement and product perception. This work proposes to combine motion capture and immersive technologies to bridge the gap between static configurators and experiential customization tools in the motorcycle industry. Future applications of this approach could redefine online configurators, making them more interactive, personalized, and engaging for users.

Index Terms: Motion capture, computer animation, visualization, real-time computer graphics, virtual reality, automotive visualization,

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tion, online 3D product configurator.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Computing methodologies—Computer graphics—Animation—Motion capture; Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools

1 INTRODUCTION AND BACKGROUND

Online configurators have become a standard in the automotive industry, allowing customers to personalize and visualize vehicles to suit their preferences [5, 6]. In the motorcycle sector, similar tools are gaining traction, enabling enthusiasts to configure bikes online [1, 20]. However, unlike car configurators, motorcycle customization is deeply intertwined with the rider's unique interaction with the vehicle. A critical aspect of creating an engaging configurator for motorcycles lies in realistically visualizing the dynamics between the motorcycle and the rider [2].

Designers and researchers often use avatars [15]—virtual characters that represent users in digital worlds—to enable direct interaction in the virtual environment [23, 11, 13, 14]. Previous research highlights that avatars enhance user identification with products, positively influencing product perception [8]. In traditional non-immersive applications, there exists a distinct separation between the digital world and the user [7, 18]. However, integrating user representation through avatars and fostering a sense of presence ("being there") can bridge this gap, providing users with a more profound and immersive product experience [18, 25, 21, 9]. Previous work found that having a body in VR and interacting through the avatar with the virtual world can increase the sense of presence [3, 22]. Consequently, incorporating a virtual rider as a visual representation within the VR experience could be a promising approach to create exciting user experiences while personalizing motorcycles [11]. This approach allows users to explore how they

would look, move, and feel on a customized motorcycle, thereby enhancing the overall experience [23, 22].

Despite the growing adoption of online 3D configurators, their focus remains primarily on configuration, often neglecting the experiential aspect. Current solutions lack accurate simulation capabilities and fail to integrate user representation effectively, limiting the potential for users to identify with the vehicle or brand. For example, a key aspect of motorcycle dynamics, such as suspension response, could be visualized realistically through avatar-based simulation, allowing users to experience how their customizations impact the vehicle's performance and feel.

Inspired by motorcycle racing video game (e.g., *MotoGP 24* or *Ride 5*) and motorcycle racing video games with VR-option (e.g., *Moto Racer 4* or *VRider SBK*), where users can customize and interact with vehicles in immersive environments, this paper proposes that such experiential features can significantly enhance online 3D configurators. This paper therefore introduces a prototype demonstrating how motion capture (MoCap) technology can be incorporated to create a realistic interaction between a motorcycle and its rider within a virtual reality (VR) application. The paper explores an optimized workflow for implementing these interactions and addresses challenges such as adapting motion data to different motorcycle models.

2 TRACKING THE MOTORCYCLE KINEMATIC

In the following, we describe a proof-of-concept prototype that proposes enhancing a motorcycle's customization experience using MoCap technology to simulate a realistic interaction between a motorcycle and the rider. In addition, this prototype aims to represent the user by providing a virtual avatar to enable a natural and embodied interaction.

2.1 Prototype Development

In collaboration with PIERER Innovation, an immersive prototype for the KTM online configurator [16] was developed. The KTM configurator is a VR-enabled application based on the Unreal Engine using Pixel Streaming. This feature allows high-quality, real-time rendering of Unreal Engine projects to be streamed to various devices (mobile devices, VR-devices, workstations, etc.) via a web browser.

PIERER Innovation provided a 350SX-F motocross bike as well as digital data. The objective was to precisely capture the forces exerted on the motorcycle during various dynamic scenarios, including mounting, dismounting, acceleration, braking, jumping, and cornering. For this purpose, the motorcycle was divided into six distinct rigid-body segments (i.e., handlebar, shock absorber/spring, frame and fairing, rear frame/suspension/swing arm, front wheel, rear wheel).

2.2 Rigid Bodies

A rigid body, as defined in this context, is a transformation derived from three or more points that are fixed relative to each other [10]. Each segment's motion is described by a combination of translational and rotational movements around a pivot point determined by the motorcycle's kinematics.

2.3 Positioning Strategy of Passive Trackers on the Motorcycle

To capture the motorcycle's kinematic movements accurately, 40 passive tracking markers, each with a diameter of 14 mm, were strategically placed on the motorcycle. These markers were mounted on base elements of varying heights to optimize placement relative to the kinematics of each segment (Fig. 2). Placement locations were selected based on their relevance to the motorcycle's motion and dynamics.

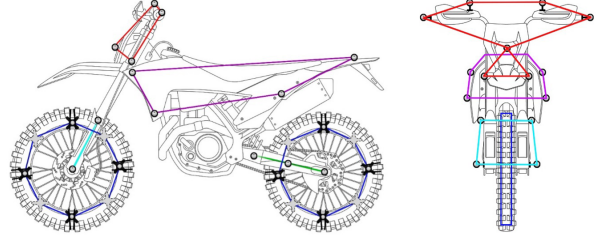


Figure 2: Division of motorcycle kinematics into six rigid-body groups. (left) side view, (right) front view.

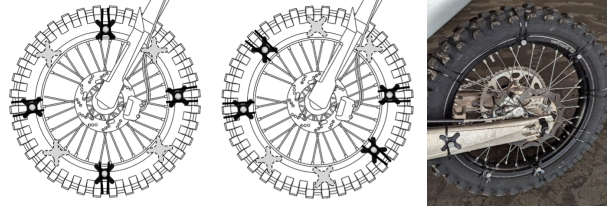


Figure 3: Different strategies for positioning tracking markers on the wheels.

2.4 Marker Placement on Wheels

The wheels presented unique challenges due to their high rotational speeds, exposure to large forces, and obstructions caused by components such as the suspension fork, swing-arm, fairing, exhaust system, and drive chain. To address these issues, eight tracking markers were tested in various configurations for each wheel (see Fig. 3). This approach ensured that the kinematic data of the wheels could be recorded with sufficient accuracy despite the mechanical constraints.

This setup provided the foundation for capturing precise motion data required for realistic simulations of motorcycle-rider interactions in a virtual environment.

2.5 Tracking The Rider

Since the MoCap software Motive (Version 3.1) [19] offers various standardized character rigs (Fig. 4), this process does not present unique challenges. We chose a character rig with 54 (14mm) passive tracking markers (Fig. 4), which tracks and records the entire body movement, including the hands but excluding individual fingers. The markers must be attached to the MoCap suit according to the rig specifications. 94 passive markers were used for the motorcycle and the rider.



Figure 4: left: Different passive tracking markers and base elements; right: Character rig in Motive;

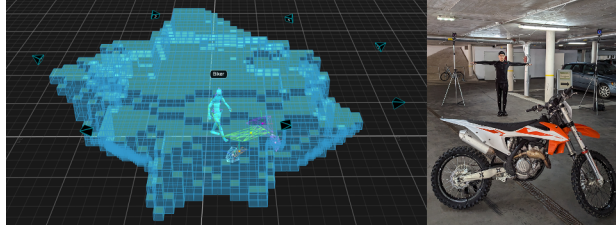


Figure 5: left: Motorcycle and rider with tracking markers; right: View of the MoCap volume in Motive.



Figure 6: left: Mobile MoCap setup; right: Perspective view of tracked elements in Motive.

2.6 Motion Capture Setup and Capturing

Eight MoCap cameras were mounted on tripods for the outdoor MoCap session. Due to the limited number of cameras, their strategic positioning was crucial for high-quality recordings. The cameras were arranged in an oval configuration of 13 x 8 meters and positioned at a height of about 2.30 meters, angled downwards at an angle of -25° to -30° toward the capture volume. Three different camera orientations towards the capture volume were tested depending on the scenario (see Figs. 7 and 8).

To improve the recording quality during interactions between the rider and the stationary motorcycle (mounting, dismounting, etc.) in the center of the capture volume, all cameras were oriented and focused on the center of the volume at approximately 0.95 meters (the motorcycle's saddle height) (see Fig. 7).

To increase the effective capture volume of the moving motorcycle, a decentralized camera orientation strategy was employed (see Fig. 8). Unlike the conventional approach of using a single focal point, two focal points were designated—one for the left side and one for the right side of the cameras, which were arranged in an oval configuration. This adjustment allowed for a slight expansion of the capture volume, accommodating a larger range of motorcycle motion. However, this modification introduced a trade-off, as the tracking quality in the center of the capture volume was marginally reduced, reflecting the limitations of the decentralized orientation approach.

2.7 Outdoor Motion Capture

Due to space constraints and fire safety regulations in the MoCap studio at the University of Applied Sciences Upper Austria, Cam-

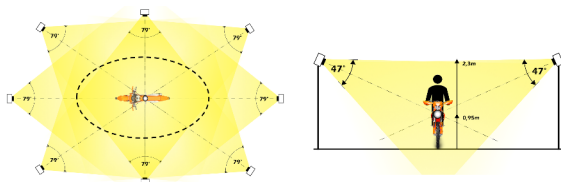


Figure 7: Orientation of MoCap cameras for stationary interaction between rider and motorcycle (top and side view).

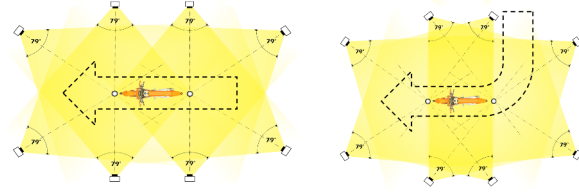


Figure 8: Decentralized orientation of the MoCap cameras for cornering and straight-line riding.

pus Hagenberg, the motorcycle scenes were recorded outdoors. Optical motion capture systems operate by illuminating passive reflective markers with infrared LED light (near-infrared range 850-940 nm). However, sunlight, which includes a full spectrum of light, can pose significant challenges. The sun's infrared spectrum can reflect off surfaces and objects, causing interference that severely impacts the tracking quality of the motion capture system.

To mitigate this issue, the experiment was conducted in a garage on the university campus. This location provided partial shade, reducing the impact of direct sunlight. Nevertheless, unavoidable stray light and reflections from pipes and parked vehicles still posed challenges. These interfering areas had to be manually masked in the MoCap software Motive.

Despite these measures, the presence of sunlight—both direct and indirect—negatively affected the overall tracking quality and range when compared to studio recordings. It is likely that selecting a location without indirect sunlight could have further improved tracking results.

Given the limited capture volume of 13x7 meters and the high speeds involved in motorcycle movement, footage was recorded at 240 frames per second (fps). To maintain focus on the reflective tracking markers throughout the capture area, cameras positioned further away in the oval setup were set to an aperture of f/5.6, while cameras positioned closer to the capture area used an aperture of f/4.0.

2.8 Cleaning Motion Capture Data

Motion capture data recorded outdoors is susceptible to inaccuracies due to marker occlusion or infrared interference caused by direct and indirect sunlight. These issues can lead to interruptions or errors in the movement paths of individual markers, making post-processing (data cleaning) necessary. This cleaning process can be performed either directly in Motive or manually using 3D software.

Correcting disrupted or missing marker movement paths requires experience in animation and a solid understanding of animation principles to reconstruct the missing motion curves accurately. However, this process is often time-consuming. To evaluate the effort and impact of data cleaning, tests were conducted on the animation both with and without correction to assess qualitative differences for KTM.

The latest version of OptiTrack Motive (Version 3.1) provides partially automated data-cleaning functions, significantly reducing the effort required for post-processing. These features include:

- *Error correction and smoothing algorithms*, which can interpolate gaps in the movement curves,
- *Detection and correction of minor marker dropouts*,
- *Enhanced realism and accuracy* of captured movement paths through automated interpolation.

These tools enable the system to address marker dropouts and minor errors effectively, reducing the need for manual adjustments.

As a result, the overall quality of the motion capture data is improved, and the cleaning process becomes both faster and more efficient. By combining careful capture strategies, such as masking and optimal camera settings, with advanced post-processing tools, the MoCap workflow achieved a balance between practical constraints and data quality, offering realistic and reliable results for the motorcycle scenes.

3 APPLICATIONS AND DISCUSSION

The integration of MoCap technology to simulate realistic interactions between a motorcycle and its rider through an avatar presents numerous opportunities across various domains. Current configurators can be enhanced by incorporating avatars and motorbike animations. For instance, the *KTM Configurator* could be expanded to include user integration, enabling personalization of avatars and providing a visual representation of how the user would look while sitting on a motorbike. Animations created by MoCap systems could simulate realistic scenarios, showcasing the dynamic interaction between the avatar and the motorbike as it moves through various street environments. In addition, immersive technologies such as Augmented Reality (AR) could be leveraged to offer an immersive experience due to their increased popularity and social acceptance [12, 24]. For example, users could visualize themselves in different outfits and also personalize the motorbike digitally projected into their own garage. Adding sounds and visual effects such as wheel spinning or suspension can contribute to the experience. Furthermore, the MoCap system that captures both the subtle and significant movements of the rider interacting with the motorcycle could lead to realistic motorcycling experiences. Players could experience the physical feel of leaning into turns, adjusting posture for acceleration, or responding to environmental factors, all represented with lifelike avatars. This level of realism has the potential to redefine user engagement and interaction within simulators and entertainment systems.

The preliminary prototype demonstrates how optical MoCap can be used to create realistic animations of the motorcycle and rider. However, these MoCap recordings are limited to a confined space and are not suitable for capturing complex maneuvers, such as cornering dynamics or high-speed evasive actions. For such intricate motion data, marker-free MoCap systems [4] or inertial-measurement-unit (IMU)-based MoCap system [17] would need to be tested and utilized.

The *KTM Configurator* offers a variety of motorcycle models with extensive customization options. Both the geometry and behavior of the motorcycles differ depending on the model, making it challenging to transfer MoCap data from one motorcycle to another. As a next step, the adaptation of animations to other motorcycle models is planned for testing. Additionally, studies on the prototype are being considered to further evaluate its performance and applications. For example, future work could systematically evaluate the realism and fidelity of motorbike animations created using MoCap compared to those developed manually. Potential target groups for such comparative evaluations might include avid motorcycle enthusiasts and professionals such as designers and animators with expertise in vehicle dynamics and physics. This would allow to cover perceptual assessments based on different perspectives.

4 CONCLUSION

This paper presents a proof-of-concept prototype that integrates MoCap technology and VR to create a more dynamic and immersive user experience for online motorcycle configurators. By incorporating realistic interactions between a motorcycle and its rider through an avatar, the prototype overcomes some of the key limitations of traditional configurators, which often fail to provide users with a sense of presence and interaction. The research demonstrates that immersive technologies, such as MoCap and VR, can enhance

user engagement and product perception, transforming the configurator experience from a static tool to an interactive, personalized platform.

The integration of MoCap to simulate realistic rider-motorcycle dynamics offers significant potential for the motorcycle industry. The ability to personalize avatars and visually represent users' interactions with the motorcycle could be further expanded by incorporating game mechanics, adding realistic sound effects, and simulating environmental factors. These advancements would not only improve configurator platforms but also provide a more holistic and engaging experience for users.

While the prototype successfully demonstrates the feasibility of MoCap in creating realistic motorcycle animations, challenges remain in capturing complex maneuvers such as cornering dynamics or high-speed actions. Future research will need to explore marker-free MoCap systems or IMU-based technologies to capture these intricate movements. Additionally, the adaptation of MoCap animations across different motorcycle models remains a critical area for further development, given the differences in geometry and behavior between models.

The next steps for this research include testing the adaptation of animations to various motorcycle models and comparing the realism of MoCap-generated animations with those created manually. These evaluations could provide valuable insights from both enthusiasts and professionals in the field, offering a comprehensive assessment of the technology's effectiveness and the user experience it provides. Overall, this work lays the foundation for future advancements in interactive configurators, with the potential to redefine how motorcycles are personalized, experienced, and engaged with in both virtual and real-world contexts.

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